Convex Optimization of Portfolio Kurtosis

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Abstract

This work presents five convex reformulations of portfolio kurtosis that allows us to pose kurtosis as a parametric convex risk measure. These new reformulations are based on new formulas for estimation of portfolio moments and co-moments matrices, second order cone and semidefinite programming. The advantages of these new reformulations are that allow us to optimize portfolio kurtosis using disciplined convex programming rules; can be solved using state of art solvers that supports second order cone and semidefinite programming; are suitable for a medium size problems; and have the flexibility to model several objective functions like risk adjusted return ratio maximization, kurtosis minimization, kurtosis constraints and risk parity based on kurtosis. Also, we present the heuristic of lower semi-kurtosis, a parametric optimization model based on lower semi-co-kurtosis matrix that allows us to minimize the left tail of portfolio returns.

Keywords: finance, convex programming, portfolio optimization, kurtosis, higher moments.

JEL Codes: C61, G11

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1 Introduction

In higher order moment portfolio optimization, portfolio kurtosis is always take as a nonconvex function like in Jurczenko et al. (2015), Niu et al. (2019) and Zhou and Palomar (2020), because traditional formula is based on a tensor formula that not fit in classic convex formulations. To overcome this problem, we present new formulas for portfolio moments that allows us to see if a particular moment can be expressed as a convex function using discipline convex programming (DCP) rules. DCP is a modeling methodology proposed in Grant and Boyd (2006), that imposes a set of conventions that one must follow when constructing convex programs. The advantage of this methodology is that the transformations necessary to convert DCP problems into solvable form can be fully automated.

This work proposes five convex formulations of portfolio kurtosis optimization using DCP rules. The first two formulations are based on semidefinite programming that are suitable only for a small number of assets . The last three formulations are a combination of semidefinite programming and second order cone programming that are suitable for portfolios with a medium number of assets. Then, we are going to show how to incorporate the convex formulation of kurtosis in problems like minimization of kurtosis, mean variance kurtosis maximization. return kurtosis ratio maximization and risk parity based on kurtosis. Also to increase the speed of the calculation we proposed a relaxed model using Kim and Kojima (2000) technique. We are going to run some numerical examples using the last kurtosis model using Python 3.9 [1], CVXPY[2] and MOSEK[3] solver; in order to show the properties of the kurtosis portfolios compared to variance portfolios. Finally, we are going to compare the performance of last kurtosis model respect to the kurtosis portfolio model based on scenarios and the relaxed model.

2 Portfolio Moments Formulas

Traditionally, portfolio moments are expressed using the following formulas:

[1] https://anaconda.org

[2] Diamond and Boyd (2016) and Agrawal et al. (2018)

[3] ApS (2022)

$$
\begin{aligned}
\mu_1 &= x' M_1 \\
\mu_2 &= x' M_2 x \\
\mu_3 &= x' M_3 (x \otimes x) \\
\mu_4 &= x' M_4 (x \otimes x \otimes x) \\
&\vdots \\
\mu_k &= x' M_k \underbrace{(x \otimes \ldots \otimes x)}_{k-1 \text{ times}}
\end{aligned} \tag{1}
$$

where μ_k is the portfolio moment of order k, M_k is the co-moment tensor of order k[4] of size $n \times n^{k-1}$, x are the portfolio weights vector of size $n \times 1$ and $\otimes$ is the kronecker product. However, using the vectorization operator vec, we can rearrange the formulas above into:

$$
\begin{aligned}
\mu_1 &= \text{vec}(M_1)' x \\
\mu_2 &= \text{vec}(M_2)' (x \otimes x) \\
\mu_3 &= \text{vec}(M_3)' (x \otimes x \otimes x) \\
\mu_4 &= \text{vec}(M_4)' (x \otimes x \otimes x \otimes x) \\
&\vdots \\
\mu_k &= \text{vec}(M_k)' \underbrace{(x \otimes \ldots \otimes x)}_{k \text{ times}}
\end{aligned} \tag{2}
$$

Also, using q-way elimination L_q matrix defined in Meijer (2005) and q-way summation matrix S_q[5], we can reduce the size of parameters used in formulas 2 and express them as:

$$
\begin{aligned}
\mu_2 &= [S_2 \text{vec}(M_2)]' [L_2 (x \otimes x)] \\
\mu_3 &= [S_3 \text{vec}(M_3)]' [L_3 (x \otimes x \otimes x)] \\
\mu_4 &= [S_4 \text{vec}(M_4)]' [L_4 (x \otimes x \otimes x \otimes x)] \\
&\vdots \\
\mu_k &= [S_k \text{vec}(M_k)]' [L_k \underbrace{(x \otimes \ldots \otimes x)}_{k \text{ times}}]
\end{aligned} \tag{3}
$$

[4] M_1 is the expected return vector, M_2 is the covariance matrix, M_3 is the co-skewness matrix and M_4 is the co-kurtosis matrix.

[5] Summation matrix S_q allow us to delete all repeated elements by indices from $\text{vec}(M_k)$ and multiply each element by the number of times it is repeated. It is defined as $S_q = D_q' D_q L_q$ where D_q is the q-way duplication matrix defined in Meijer (2005).

Also, we can express portfolio moments of order $2k$ as a quadratic form or sum of squares (SOS) formulation using the following formulas:

$$\begin{aligned} \mu_2 &= x'\Sigma_2 x \\ \mu_4 &= (x \otimes x)'\Sigma_4(x \otimes x) \\ &\vdots \\ \mu_{2k} &= \underbrace{(x \otimes \ldots \otimes x)'}_{k\,\text{times}} \Sigma_{2k} \underbrace{(x \otimes \ldots \otimes x)}_{k\,\text{times}} \end{aligned} \tag{4}$$

where Σ_{2k} is a stacked version of M_{2k}[6] and has size $n^k \times n^k$. In appendix A we propose some matricial formulas to directly estimate population co-moments tensors M_k and matrixes Σ_{2k}.

In the case of kurtosis, we have two additional formulas that allow us to take advantage of the symmetry of $(x \otimes x \otimes x \otimes x)$ and $(x \otimes x)$, and using the duplication summation matrix S_2 that we proposed and duplication elimination matrix L_2 defined in Magnus and Neudecker (1980), to reduce the number of parameters of kurtosis formulas 2 and 4:

$$\mu_4 = [(S_2 \otimes S_2)\text{vec}(M_4)]' \, [(L_2 \otimes L_2)(x \otimes x \otimes x \otimes x)] \tag{5}$$

$$\mu_4 = [L_2(x \otimes x)]' \, [S_2 \Sigma_4 S_2'] \, [L_2(x \otimes x)] \tag{6}$$

3 Convex Kurtosis Portfolio Optimization

3.1 Kurtosis Minimization

The minimization of portfolio kurtosis is usually considered as a nonconvex problem, this problem can be posed as:

[6] Using Python package Numpy we can get `Sigma_2k = numpy.reshape(M_2k, (n**k, n**k), order='F')`

$$
\begin{aligned}
&\min_{x} x'M_4(x \otimes x \otimes x) \\
&s.t. \sum_{i=1}^{n} x_i = 1 \\
&x \geq 0
\end{aligned}
\tag{7}
$$

If we replace the objective function with kurtosis formula 2 we get the following problem:

$$
\begin{aligned}
&\min_{x} \operatorname{vec}(M_4)'\operatorname{vec}(x \otimes x \otimes x \otimes x) \\
&s.t. \sum_{i=1}^{n} x_i = 1 \\
&x \geq 0
\end{aligned}
\tag{8}
$$

We can noticed that $x \otimes x = \operatorname{vec}(xx')$, so we can reformulate the problem above introducing the variables $X \in \mathbf{S}^n$[7] and $Z \in \mathbf{S}^n$, and constraints $X = xx'$ and $Z = zz'$. These constraints are equivalent to the Schur complements constraints $X - xx' \succeq 0$ and $Z - zz' \succeq 0$; and rank constraints $\mathbf{rank}(X) = 1$ and $\mathbf{rank}(Z) = 1$. We can notice that the constraints $\mathbf{rank}(X) = 1$ and $\mathbf{rank}(Z) = 1$ are not DCP, so we can relax the problem and make it DCP ignoring these constraints as follows:

$$
\begin{aligned}
\min_{x,X,z,Z} \quad & \operatorname{vec}(M_4)'\operatorname{vec}(Z) \\
& \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& z = \operatorname{vec}(X) \\
& \begin{bmatrix} Z & z \\ z' & 1 \end{bmatrix} \succeq 0 \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \in \mathbf{S}^n \\
& Z \in \mathbf{S}^{n^2}
\end{aligned}
\tag{9}
$$

In this case, the size of Z is (n^2, n^2). However, we can reduce the size of Z to

[7] $\mathbf{S}^n$ is the set of symmetric matrices and is defined as $\mathbf{S}^n = \{X \in \mathbf{R}^{n \times n} \mid X = X'\}$

$\left(\frac{n(n+1)}{2}, \frac{n(n+1)}{2}\right)$ using kurtosis formula 5 as follows:

$$\begin{aligned}
\min_{x,\,X,\,z,\,Z}\ & [(S_2 \otimes S_2)\mathrm{vec}(M_4)]' \,\mathrm{vec}(Z) \\
& \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& z = L_2 \mathrm{vec}(X) \\
& \begin{bmatrix} Z & z \\ z' & 1 \end{bmatrix} \succeq 0 \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \in \mathbf{S}^n \\
& Z \in \mathbf{S}^{n(n+1)/2}
\end{aligned} \tag{10}$$

But problems based on chained semidefinite constraints like 9 and 10 are difficult to solve and not scale well when we increase the number of assets. On the other hand, we can take advantage of kurtosis formula 4 to reformulate kurtosis minimization as a quadratic problem with a semidefinite constraint:

$$\begin{aligned}
\min_{x,\,X,\,z}\ & z'\Sigma_4 z \\
& \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& z = \mathrm{vec}(X) \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \in \mathbf{S}^n
\end{aligned} \tag{11}$$

We can notice from formulation 11 that a requirement to minimize kurtosis is that matrix Σ_4 must be positive semidefinite. Additionally, we can improve the performance of this model using second order cone programming to reformulate the objective function:

$$
\begin{aligned}
\min_{x,X,z,g}\ & g \\
& \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& z = \text{vec}(X) \\
& \left\| \Sigma_4^{1/2} z \right\|_2 \leq g \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \in \mathbf{S}^n
\end{aligned} \tag{12}
$$

where $\Sigma_4^{1/2}$ is the square root of matrix Σ_4, g is the square root of portfolio kurtosis and $\|\,.\,\|_2$ is the ℓ^2 norm or euclidean norm. On the other hand, variable z in problem 12 has size $n^2 \times 1$, but using kurtosis formula 6, we can reduce the size of z to $\left(\frac{n(n+1)}{2}, 1\right)$ and pose the minimization of kurtosis as follows:

$$
\begin{aligned}
\min_{x,X,z,g}\ & g \\
& \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& z = L_2 \text{vec}(X) \\
& \left\| [S_2 \Sigma_4 S_2']^{1/2} \, z \right\|_2 \leq g \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \in \mathbf{S}^n
\end{aligned} \tag{13}
$$

Due to second order cone constraint, problems 11, 12 and 13 scale better when we increase the number of assets. However, due to the semidefinite constraint this formulations still have problems when we increase the number of assets. To improve the performance of last model we are going to use the second order cone relaxation of semidefinite constraints proposed by Kim and Kojima (2000) getting the following relaxed problem:

$$
\begin{aligned}
\min_{x,X,z,g} \; & g \\
& \left\| \begin{array}{c} 1 - C_i \bullet X \\ 2L_i' x \end{array} \right\| \leq 1 + C_i \bullet X \, , \;\; \forall i \in \{1, 2, \ldots, \ell\} \\
& z = L_2 \text{vec}(X) \\
& \left\| \left[S_2 \Sigma_4 S_2' \right]^{1/2} z \right\|_2 \leq g \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \succeq 0 \\
& X \in \mathbf{S}^n
\end{aligned}
\tag{14}
$$

where $\bullet$ is the inner product operator and C_i is an arbitrary semidefinite positive matrix that has a Cholesky decomposition $C_i = L_i L_i'$[8].

Finally, we can notice that all formulations presented in this section let us to use the heuristic of lower semi kurtosis matrix, that allows to minimize the left tail of portfolio returns distribution only replacing Σ_4 for the lower semi kurtosis co-moment matrix[9].

3.2 Mean Variance Kurtosis Portfolio

Using the convex reformulation of kurtosis proposed in the section above, we can posed the mean variance kurtosis portfolio as follows:

[8]Our experiments suggest that a good choice is make $\ell = 1$ and $C_1 = M_2$

[9]In appendix A we show how to calculate the lower semi kurtosis matrix with matricial formulas

$$
\begin{aligned}
\min_{x,\, X,\, z,\, Z}\ & M_1 x - \lambda_1 \sigma - \lambda_2 g \\
& \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& z = L_2 \text{vec}(X) \\
& \left\| [S_2 \Sigma_4 S_2']^{1/2} \, z \right\|_2 \leq g \\
& \left\| \Sigma_2^{1/2} x \right\|_2 \leq \sigma \\
& \sum_{i=1}^{n} x_i = 1 \\
& x \geq 0 \\
& X \in \mathbf{S}^n
\end{aligned}
\tag{15}
$$

where σ is the portfolio standard deviation, λ_1 and λ_2 are risk aversion coefficients.

3.3 Risk Adjusted Return Portfolio

In this case, due to kurtosis is a four degree function that has a second order cone representation, we can maximize the ratio return square root kurtosis solving the following problem:

$$
\begin{aligned}
\min_{y,\, Y,\, z,\, k}\ & g \\
& \begin{bmatrix} Y & y \\ y' & k \end{bmatrix} \succeq 0 \\
& z = L_2 \text{vec}(Y) \\
& \left\| [S_2 \Sigma_4 S_2']^{1/2} \, z \right\|_2 \leq g \\
& M_1 y - r_f k = 1 \\
& \sum_{i=1}^{n} y_i = k \\
& y \geq 0 \\
& Y \in \mathbf{S}^n
\end{aligned}
\tag{16}
$$

where k is an auxiliary variable, r_f is the risk free rate and the optimal portfolio is obtained making the transformation $x = y / \sum_{i=1}^{n} y_i$.

3.4 Kurtosis Risk Parity Portfolio

We can not use a risk parity approach directly to kurtosis because it is not a homogeneous function of degree 1, however we can use this approach to the fourth root of kurtosis because it is a homogeneous function of degree 1. From problem 13 we can see that kurtosis allows a risk parity constraint based on variable z and g, but to apply the risk contribution constraint to x we can solve the following problem:

$$
\begin{aligned}
\min_{y,\,Y,\,z,\,k} \; & g \\
& \begin{bmatrix} Y & y \\ y' & 1 \end{bmatrix} \succeq 0 \\
& z = L_2 \text{vec}(Y) \\
& \left\| \left[S_2 \Sigma_4 S_2' \right]^{1/2} z \right\|_2 \leq g \\
& \sum_{i=1}^{n(n+1)/2} b_i^z \ln(z_i) \geq 1 \\
& b^z = S_2 \left[b^y \otimes b^y \right] \\
& \sum_{i=1}^{n} y_i = k \\
& y \geq 0 \\
& Y \in \mathbf{S}^n
\end{aligned}
\tag{17}
$$

where k is an auxiliary variable, b^z is the risk contribution constraint vector of variable z, b^y is the risk contribution constraint vector respect to portfolio weights y and the optimal portfolio is obtained making the transformation $x = y / \sum_{i=1}^{n} y_i$.

4 Numerical Examples

We select 30 assets (i.e., stocks JCI, TGT, CMCSA, CPB, MO, T, APA, MMC, JPM, ZION, PSA, BAX, BMY, AAPL, PCAR, BA, TMO, TXT, DE, MSFT, HPQ, SEE, VZ, CNP, NI, JNJ, PFE, AMZN, GE and GOOG) from the S&P 500 (NYSE) and download daily adjusted closed prices from Yahoo Finance for the period from January 1, 2016 to December 30, 2021. Then, we calculated daily returns building a returns matrix of size $T = 1509$ and $N = 30$. To calculate the portfolios we use Python 3.9, CVXPY and MOSEK solver.

4.1 Minimization of Kurtosis

In this section, we are going to compare two portfolios: minimum variance and minimum kurtosis.

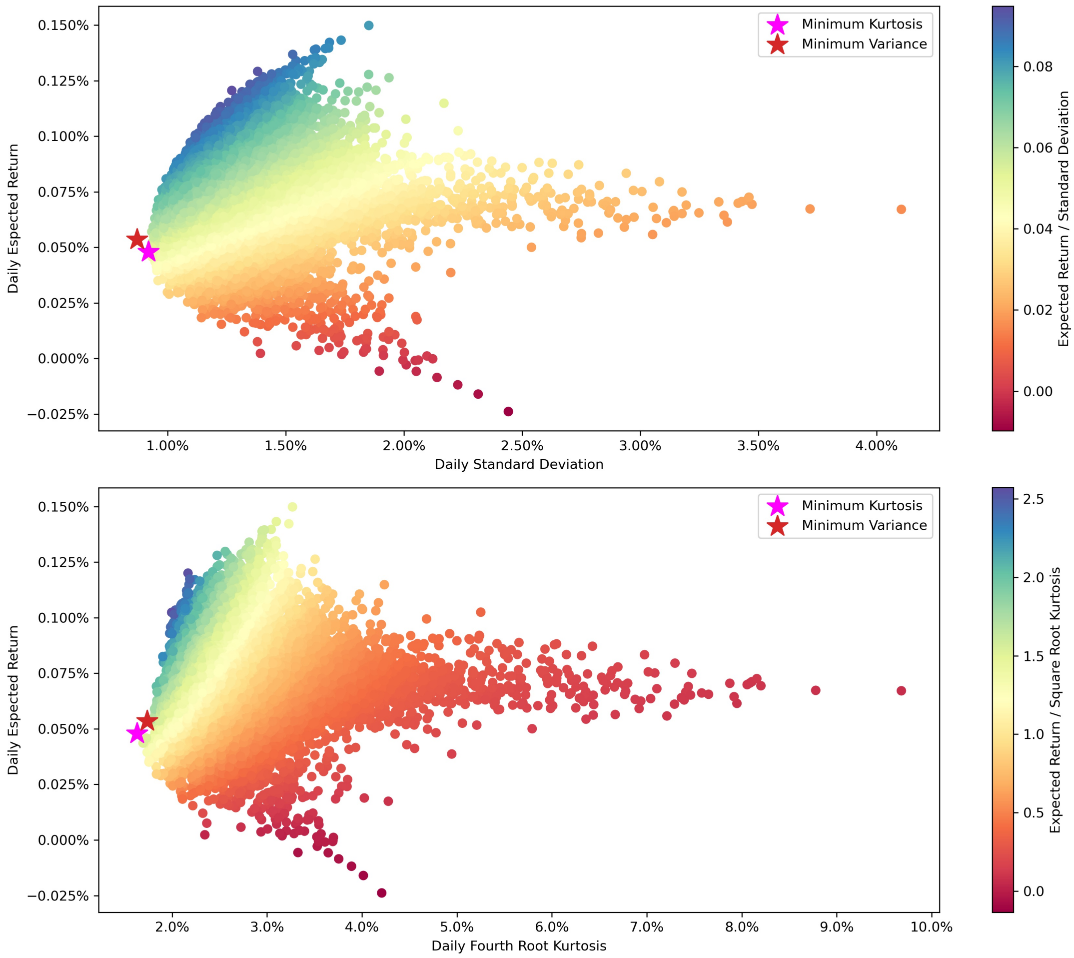

Figure 1: Optimal Portfolios in Efficient Frontier

In figure 1 we can see that both portfolios are close, but the new model achieve the goal to minimize kurtosis but has a higher variance than minimum variance portfolio.

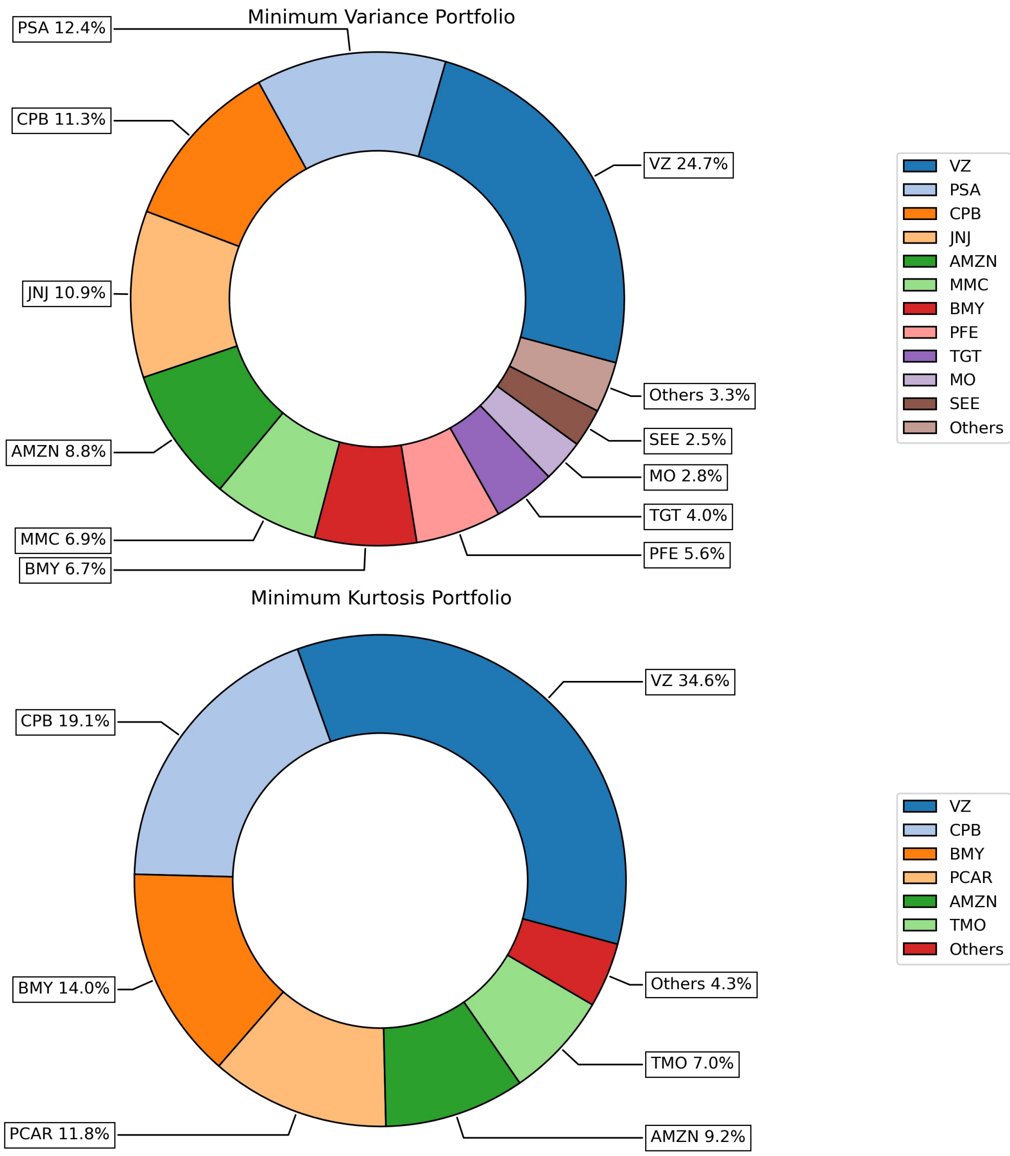

Figure 2: Optimal Portfolios Composition

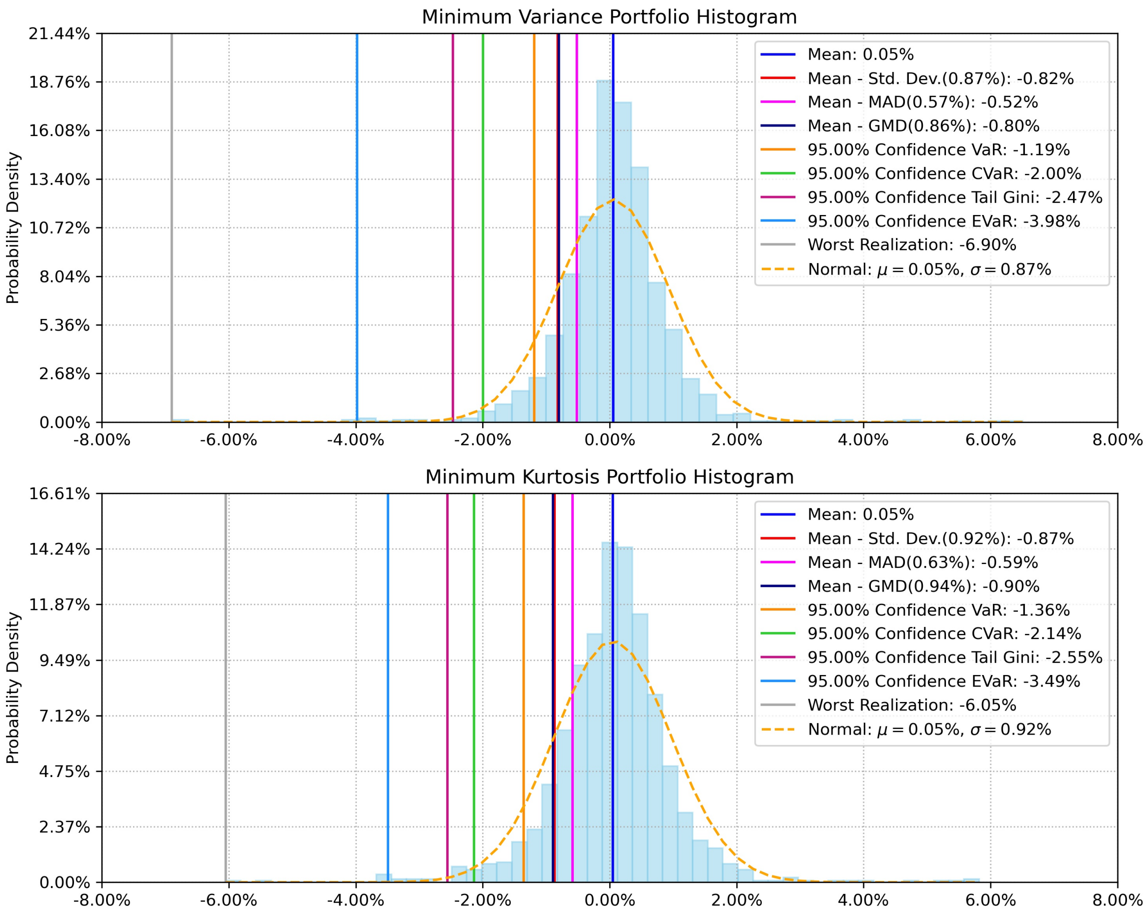

Figure 3: Optimal Portfolios Histogram

In figure 2 we can see that both portfolios have very different compositions, this means that the objective to minimize kurtosis is totally different than minimize variance. Also in figure 3 we can see that the minimum kurtosis portfolio creates a portfolio with thinner tails because it has lower entropic value at risk (EVaR) and worst realization.

4.2 Maximization of Mean Variance Kurtosis

In this section, we are going to compare two portfolios: mean variance and mean variance kurtosis with risk aversion factors $\lambda_1 = 0.5$ and $\lambda_2 = 0.25$.

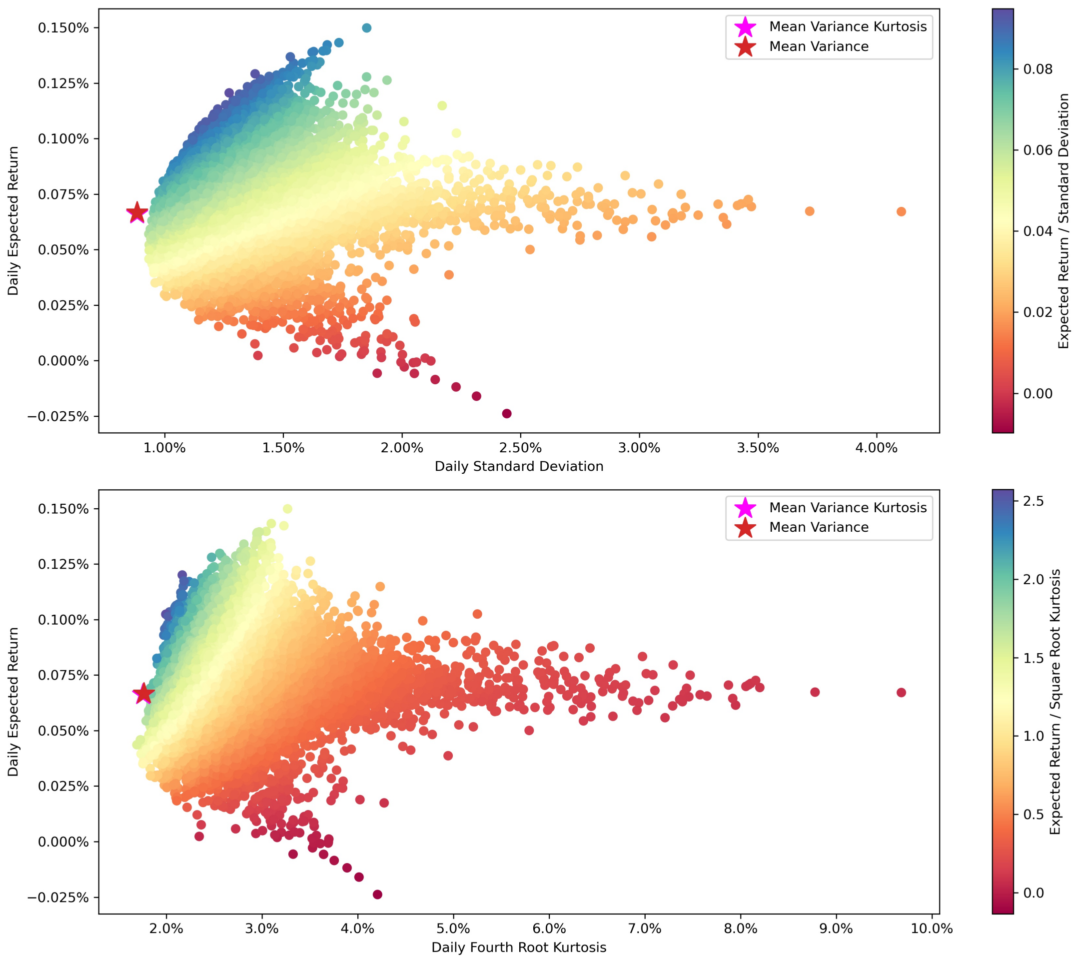

Figure 4: Optimal Portfolios in Efficient Frontier

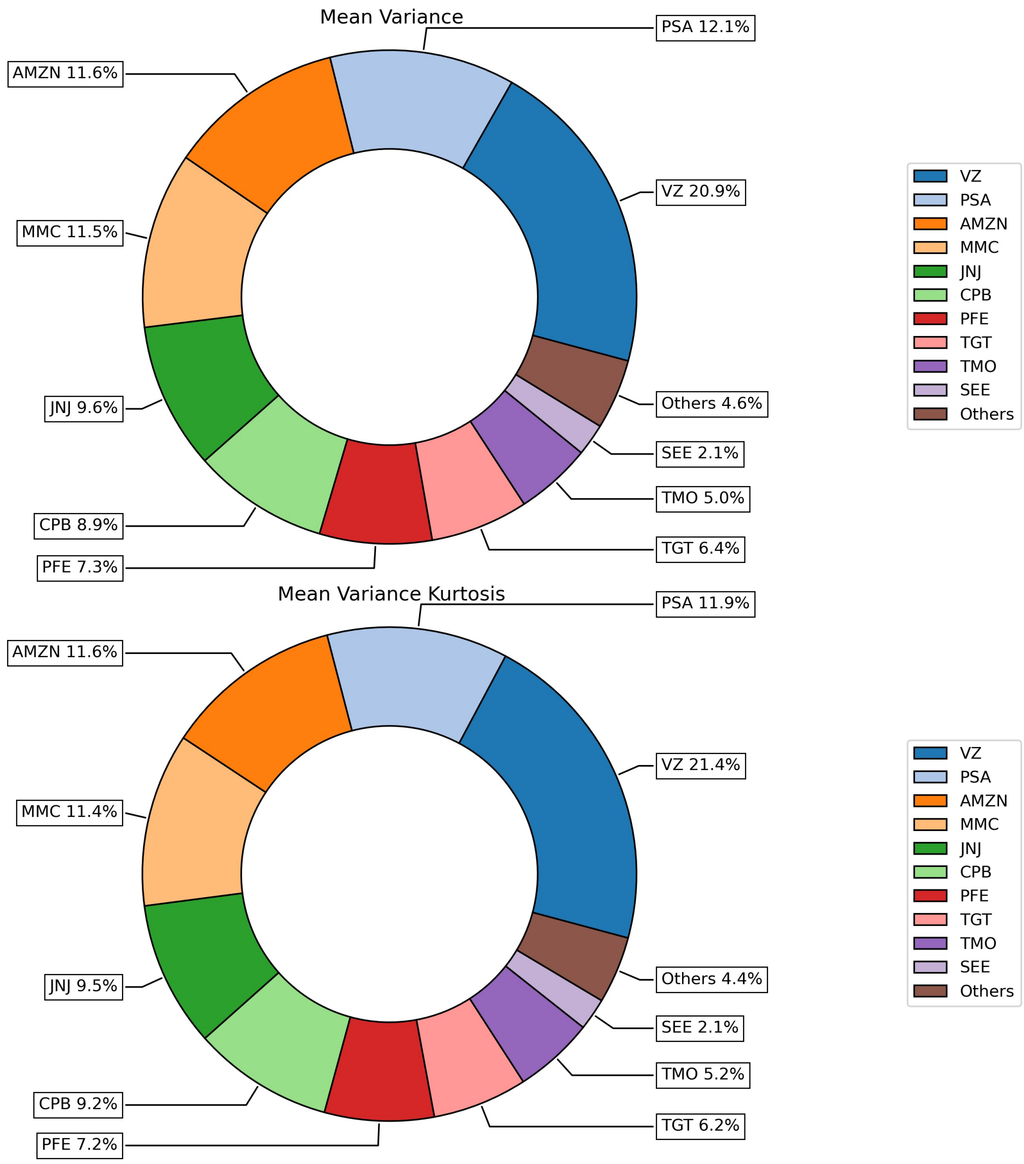

Figure 5: Optimal Portfolios Composition

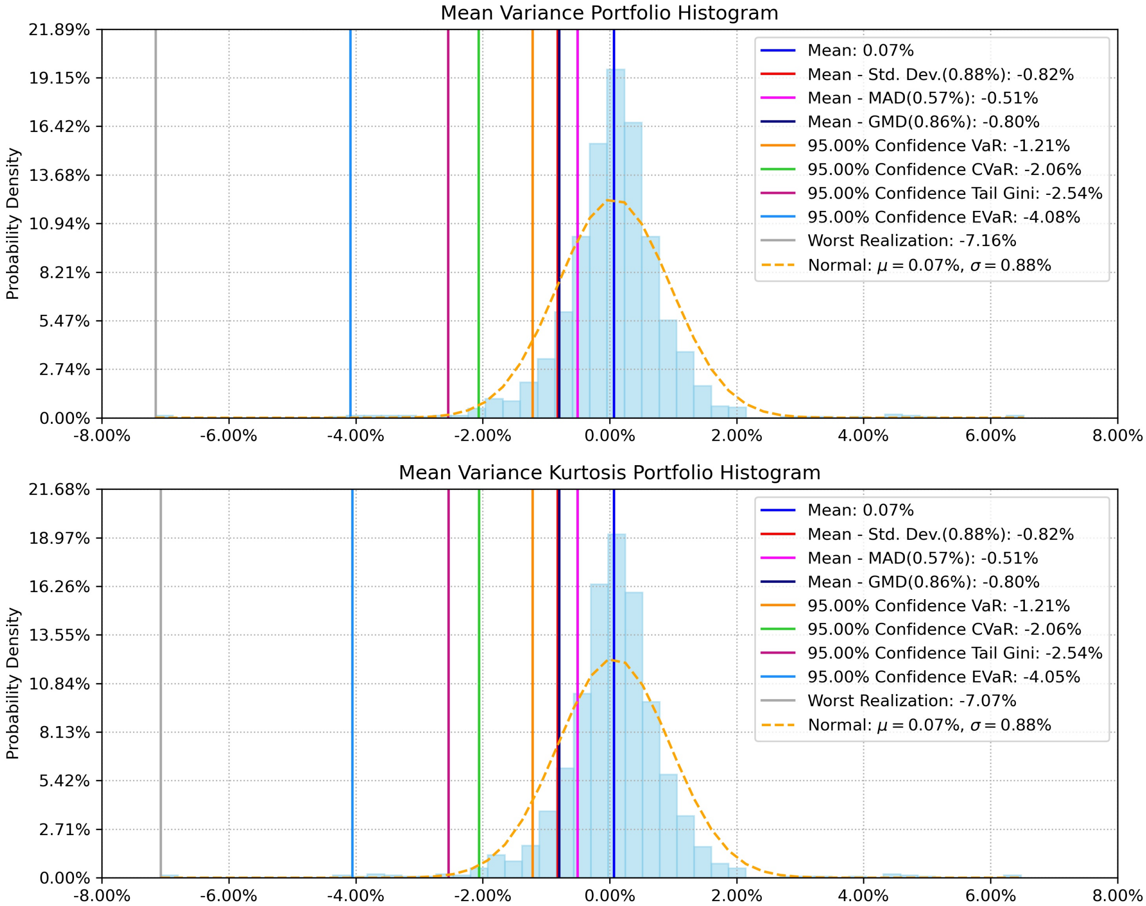

Figure 6: Optimal Portfolios Histogram

In figure 4 we can see that both portfolios are very close in the return-standard deviation and return-kurtosis planes. Also, in figure 5 we can see that both portfolios have similar composition, this means that the objective function in both cases are very similar. However, in figure 6 we can see that the inclusion of kurtosis in the objective function makes that the optimal portfolio has thinner tails because it has lower EVaR and worst realization.

4.3 Maximization of Return Kurtosis Ratio

In this section, we are going to compare two portfolios: max mean variance ratio and max mean kurtosis.

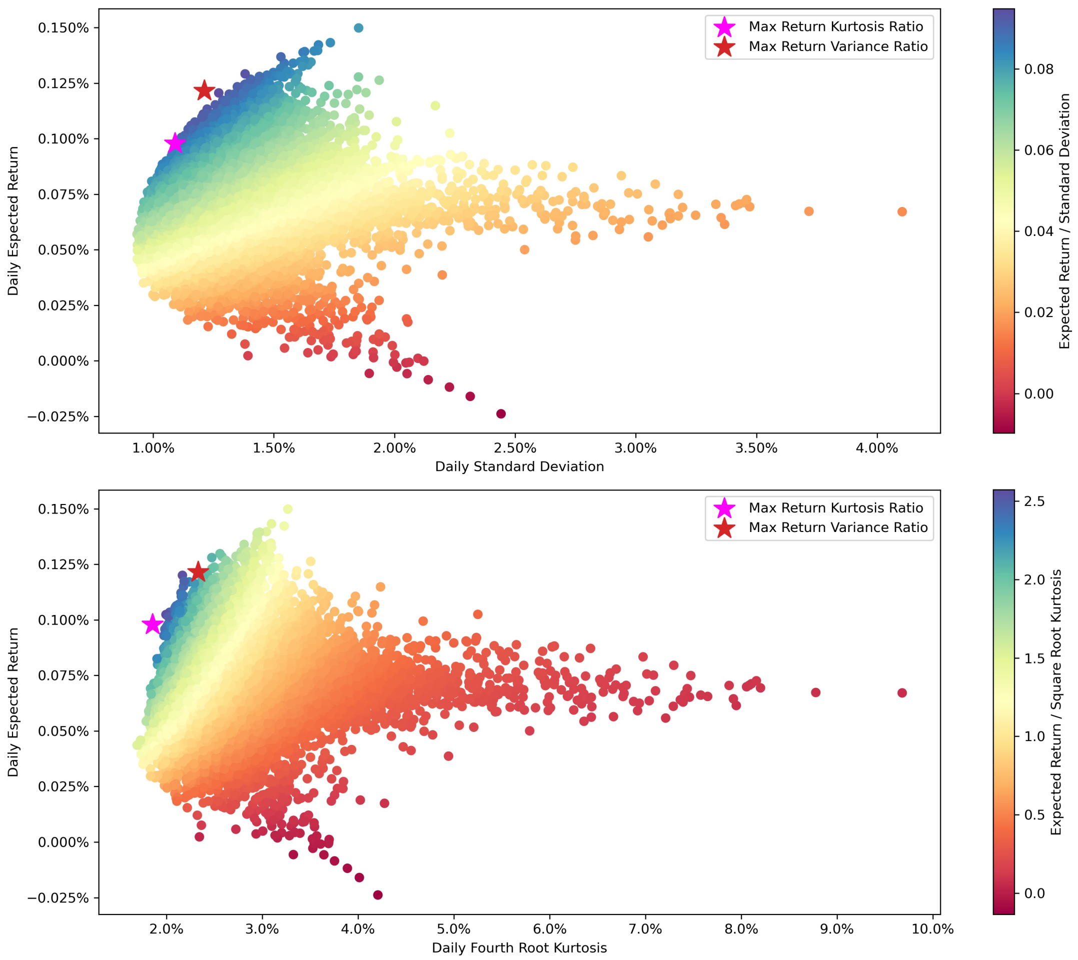

Figure 7: Optimal Portfolios in Efficient Frontier

In figure 7 we can see that both portfolios are far away because the risk adjusted return ratios have a different behavior when risk measure is standard deviation or when is square root kurtosis.

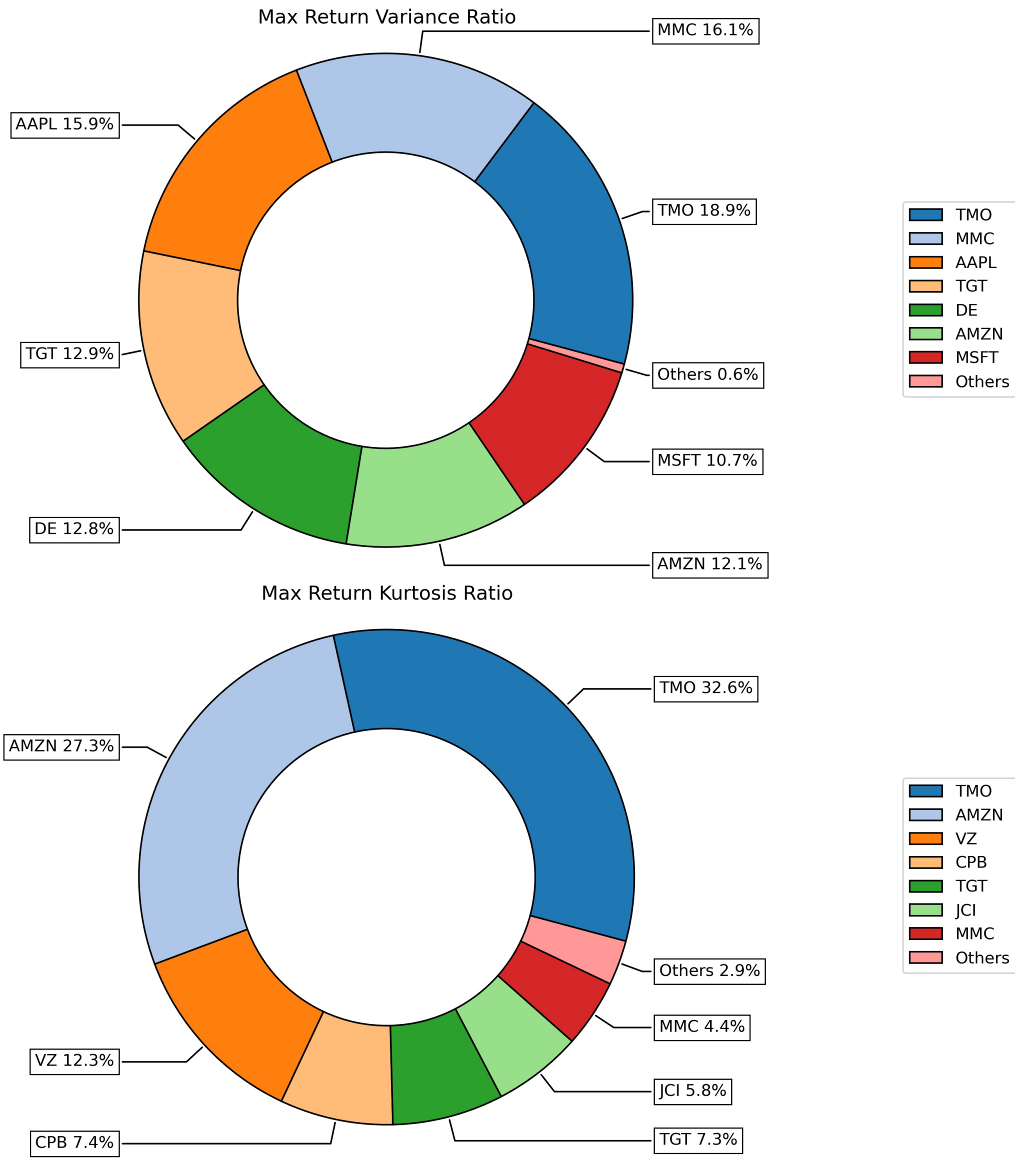

Figure 8: Optimal Portfolios Composition

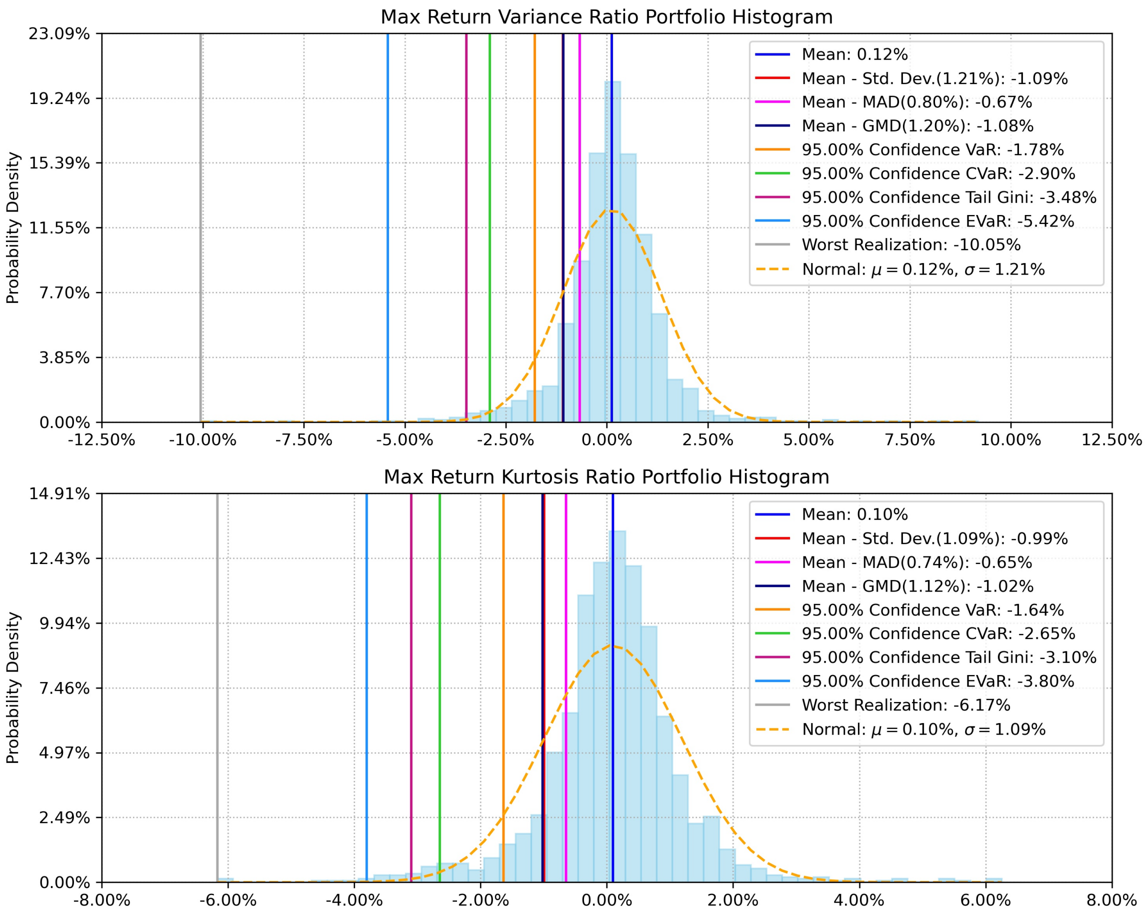

Figure 9: Optimal Portfolios Histogram

In figure 8 we can see that both portfolios have very different composition, this means that the objective function in both cases are very different. On the other hand, in figure 9 we can see that the objective function that maximizes the mean kurtosis ratio makes that the optimal portfolio has thinner tails because it has lower EVaR and worst realization.

4.4 Risk Parity Based on Kurtosis

In figure 10, we can see how the equal risk contribution portfolio effectively split the fourth root of kurtosis among all assets. Additionally in figure 11, we can see how this formulation also allows us to use custom risk contribution constraints in the risk parity problem.

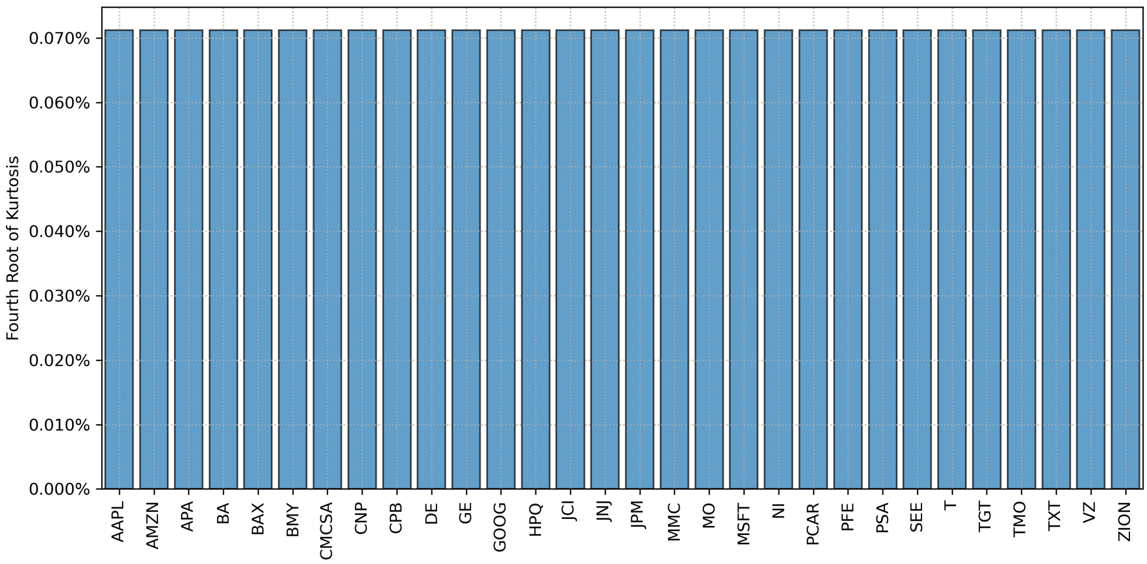

Figure 10: Risk Contribution of Equal Risk Contribution Portfolio

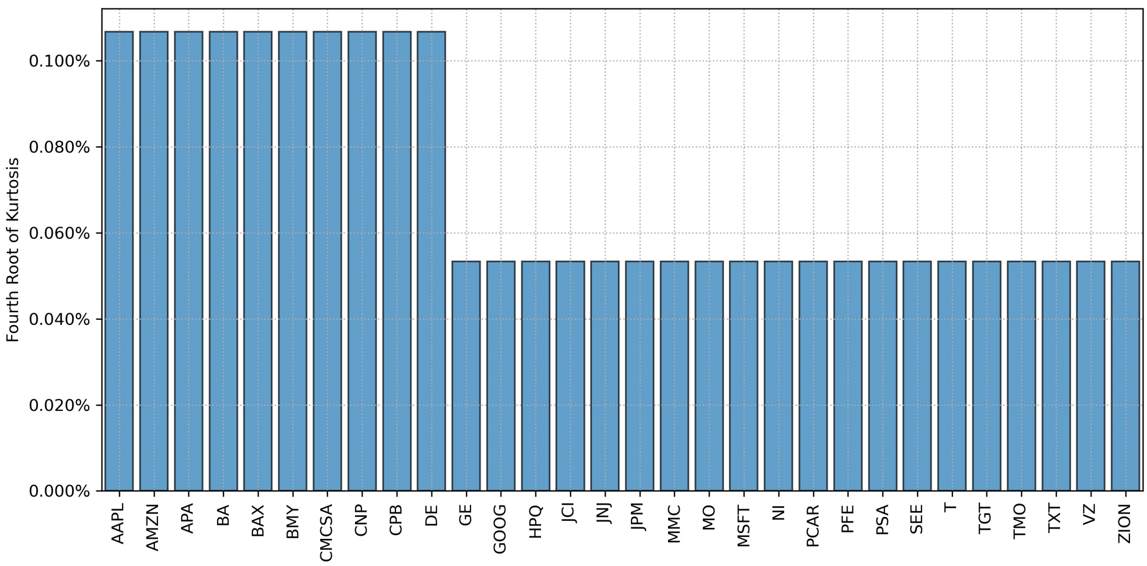

Figure 11: Risk Contribution of Custom Risk Contribution Portfolio

4.5 Semi-Kurtosis Minimization

In this section, we are going to compare two portfolios: minimum semi variance based on scenarios and minimum semi kurtosis heuristic.

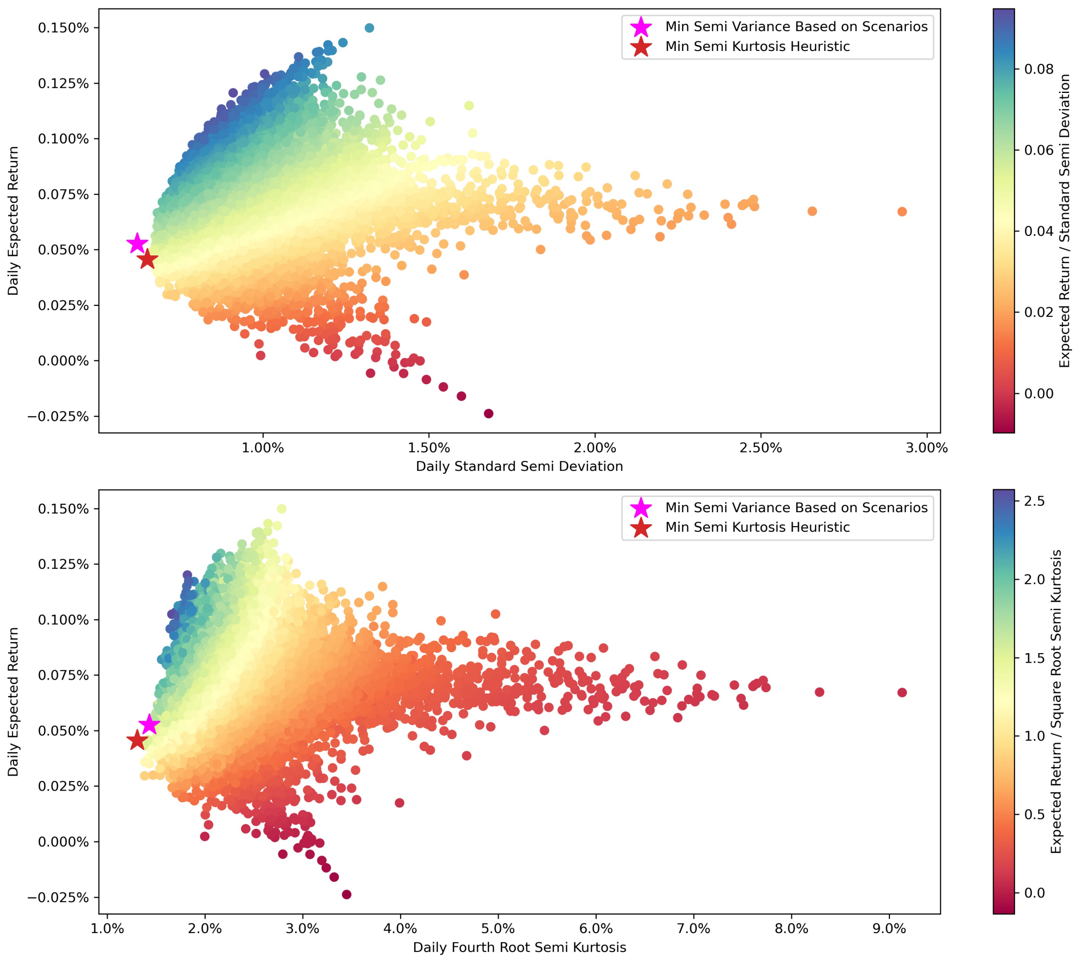

Figure 12: Optimal Portfolios in Efficient Frontier

In figure 12 we can see that both portfolios are close, but the new model achieve the goal to minimize semi kurtosis but has a higher semi variance than minimum semi variance portfolio.

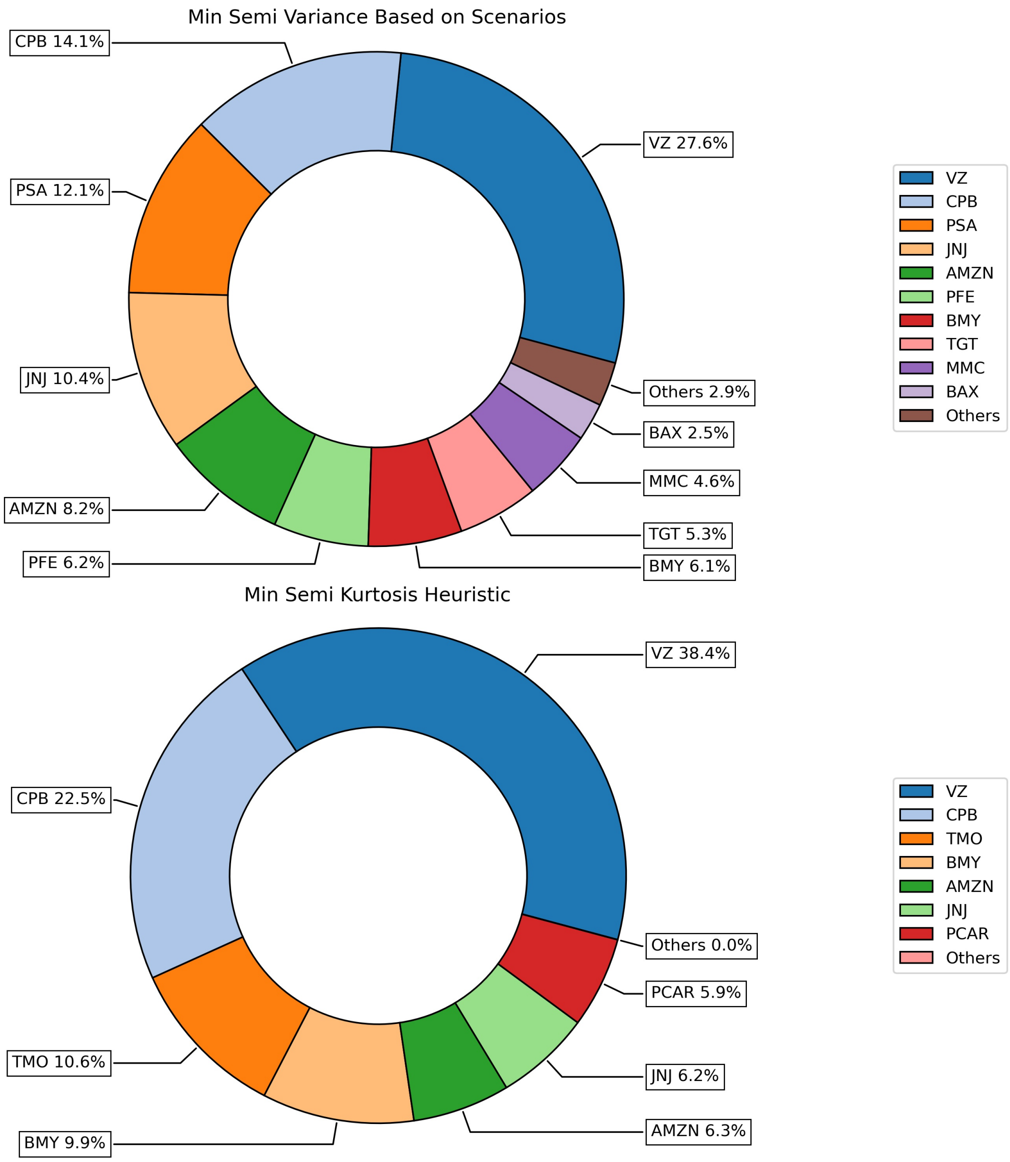

Figure 13: Optimal Portfolios Composition

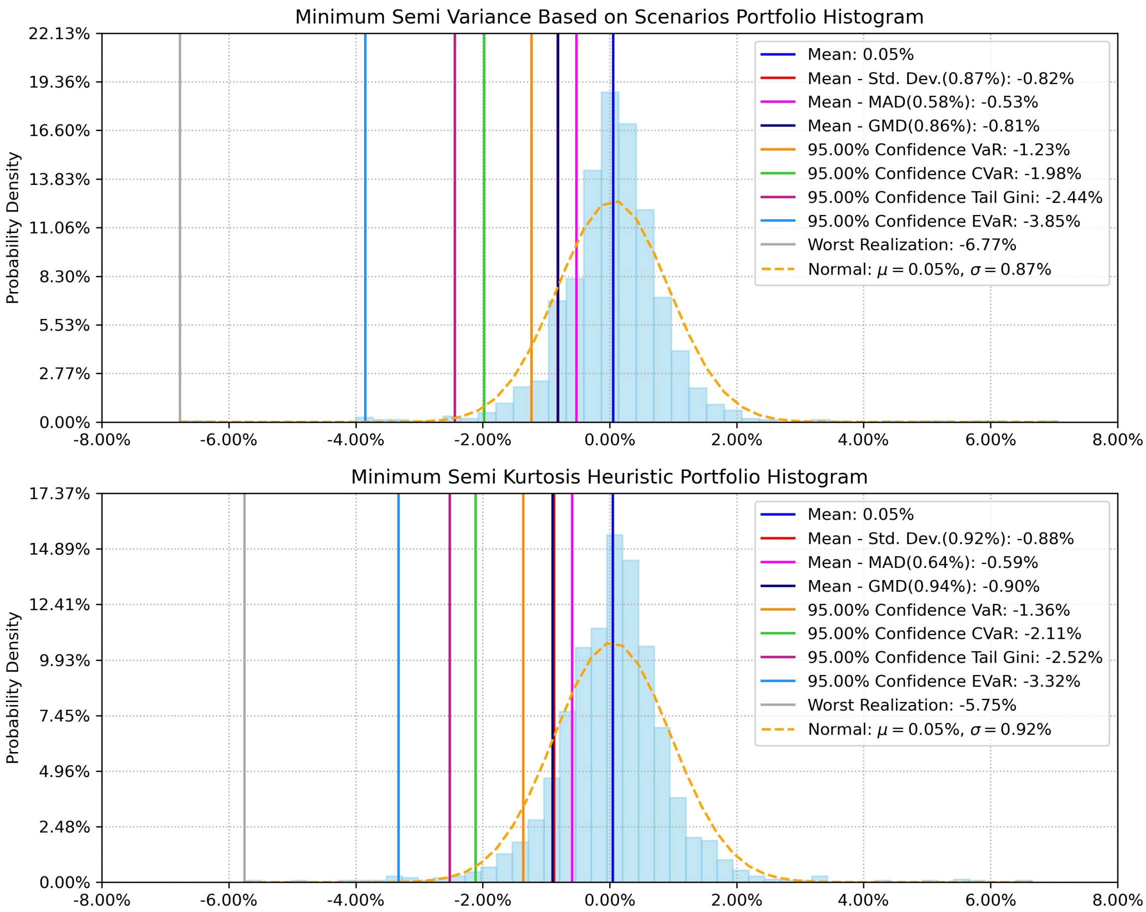

Figure 14: Optimal Portfolios Histogram

In figure 13 we can see that both portfolios have very different compositions, this means that the objective to minimize semi kurtosis is totally different than minimize semi variance. Also in figure 14 we can see that the minimum semi kurtosis portfolio creates a portfolio with thinner tails because it has lower entropic value at risk (EVaR) and worst realization.

5 Performance of Kurtosis Portfolios

In this section we compare the performance of parametric, scenario[10] and relaxation kurtosis models. We simulated 10,000 and 20,000 normal returns for 10, 30, 50 and 70 assets. Then, we estimated 3 instances of the portfolios that maximizes three objective functions: minimization of risk, maximization of risk adjusted return and risk parity for kurtosis. Then, we computed the average time and average value for each combination of sample size, objective function and risk measure. The simulation was computed

[10]These models are shown in appendix B

using a computer with two core 1.8-GHz processors and 4 GB of RAM, Python 3.9, CVXPY package and MOSEK solver.

Objective	Sample Size	Assets	Parametric (1)	Scenario (2)	Relaxation (3)	(1)-(2)	(1)-(3)
MinRisk	10000	10	0.11	3.59	0.05	-3.48	0.06
		30	1.52	4.96	0.55	-3.44	0.97
		50	12.62	6.44	5.23	6.18	7.39
		70	65.71	7.85	27.78	57.86	37.93
	20000	10	0.11	7.53	0.05	-7.42	0.06
		30	1.51	10.43	0.5	-8.92	1.01
		50	11.65	12.81	4.83	-1.16	6.82
		70	64.34	15.89	26.71	48.45	37.63
Risk Adjusted Return Ratio	10000	10	0.11	3.67	0.06	-3.56	0.05
		30	1.81	5.73	0.64	-3.92	1.17
		50	12.86	6.35	6.16	6.51	6.7
		70	61.62	7.98	30.7	53.64	30.92
	20000	10	0.11	7.51	0.06	-7.4	0.05
		30	1.6	10.46	0.57	-8.86	1.03
		50	11.51	13.62	5.39	-2.11	6.12
		70	54.58	16.66	30.06	37.92	24.52
RiskParity	10000	10	0.14	3.65	0.07	-3.51	0.07
		30	1.76	5.21	0.64	-3.45	1.12
		50	13.87	5.99	5.67	7.88	8.2
		70	73.82	7.6	27.35	66.22	46.47
	20000	10	0.15	7.27	0.07	-7.12	0.08
		30	1.76	10.27	0.67	-8.51	1.09
		50	13.54	12.32	5.51	1.22	8.03
		70	71.47	15.51	29.18	55.96	42.29

Table 1: Average Run Time in Seconds of Kurtosis Portfolios

We can see in Table 1 that average run time of parametric model are lower than scenario kurtosis for a 10 and 30 assets. In the case of 50 assets the average run time is higher for 10,000 sample size and lower for 20,000 sample size. On the other hand, for 70 assets, the average run time of the parametric model is very high compared to scenario model due to the semidefinite constraint. In the case of relaxation model, we can see that in all cases the relaxation model takes about the half of time of parametric model.

We can see in Table 2 that average objective value of parametric model are the same of scenario model for minimization of risk and risk parity objective functions. In the case of maximization of risk adjusted return, we can see that the parametric model gives a higher value of the objective function compared to the scenario model because the parametric model has an objective function closer to maximization return kurtosis ratio. Parametric model maximizes return square root kurtosis ratio while scenario model maximizes return fourth root kurtosis ratio. In the case of relaxation model,

Objective	Sample Size	Assets	Parametric (1)	Scenario (2)	Relaxation (3)	(1)-(2)	(1)-(3)
MinRisk	10000	10	0.000963	0.000963	0.000963	0	0
		30	0.001727	0.001727	0.001729	0	-0.000002
		50	0.002983	0.002983	0.002986	0	-0.000003
		70	0.00464	0.00464	0.004645	0	-0.000005
	20000	10	0.001112	0.001112	0.001112	0	0
		30	0.001565	0.001565	0.001565	0	0
		50	0.003643	0.003643	0.003645	0	-0.000002
		70	0.004957	0.004957	0.004963	0	-0.000006
Risk Adjusted Return Ratio	10000	10	0.753081	0.734292	0.752902	0.018789	0.000179
		30	0.551065	0.506304	0.550569	0.044761	0.000496
		50	0.313815	0.264055	0.313476	0.04976	0.000339
		70	0.394972	0.365139	0.394635	0.029833	0.000337
	20000	10	0.739622	0.711125	0.739534	0.028497	0.000088
		30	0.770436	0.707483	0.770203	0.062953	0.000233
		50	0.34047	0.316852	0.340378	0.023618	0.000092
		70	0.224553	0.212068	0.224454	0.012485	0.000099
RiskParity	10000	10	0.001485	0.001485	0.001526	0	-0.000041
		30	0.002748	0.002748	0.002911	0	-0.000163
		50	0.005226	0.005226	0.00543	0	-0.000204
		70	0.006996	0.006996	0.0071	0	-0.000104
	20000	10	0.001363	0.001363	0.001381	0	-0.000018
		30	0.002434	0.002434	0.002567	0	-0.000133
		50	0.005212	0.005212	0.005339	0	-0.000127
		70	0.007043	0.007043	0.007167	0	-0.000124

Table 2: Average Objective Value of Kurtosis Portfolios

we can see that in all cases the relaxation model gives close objective values to the parametric model.

6 Conclusions

The proposed convex reformulation of portfolio kurtosis allows us to incorporate kurtosis as a risk measure in the asset allocation process. This formulation has the flexibility to allow us to pose several problems like kurtosis constraints, kurtosis minimization, risk adjusted return ratio maximization and risk parity based on kurtosis. Also, due to this model relies in a semidefinite constraint and a second order cone constraint, it scale better than the formulation based only on semidefinite programming constraints for medium size problems. Another advantage is that this formulation can be solved using disciplined convex programming softwares that supports second order cone programming and semidefinite programming. In addition, this parametric formulation allows us to use the heuristic of lower semi kurtosis matrix, that minimizes the left tail of portfolio returns distribution. Also, to reduce the computation time, we proposed a model that relax the semidefinite constraint to a second order cone constraint. Finally, this formulation makes portfolio optimization based on kurtosis accessible to most users using state of art solvers.

A Matricial Formulas for Population Co-moments Matrixes

Traditionally population co-moments matrixes M_k of a returns matrix R of size $T \times n$ for $k \geq 3$ are calculated using loops. However, to increase the speed of the calculation of these parameters we propose the following matricial formulas:

$$
\begin{aligned}
M_2 &= \frac{1}{T}\left[Z'Z\right] \\
M_3 &= \frac{1}{T}\left[Z'\left((Z \otimes \mathbf{1}) \odot (\mathbf{1} \otimes Z)\right)\right] \\
M_4 &= \frac{1}{T}\left[Z'\left((Z \otimes \mathbf{1} \otimes \mathbf{1}) \odot (\mathbf{1} \otimes Z \otimes \mathbf{1}) \odot (\mathbf{1} \otimes \mathbf{1} \otimes Z)\right)\right] \\
&\;\;\vdots \\
M_k &= \frac{1}{T}\left[Z'\left(\bigodot_{i=1}^{k-1} W_i^{k-1}\right)\right] \\
W_i^s &= \underbrace{\mathbf{1} \otimes \mathbf{1} \otimes \ldots \otimes \underbrace{Z}_{\text{position } i} \otimes \ldots \otimes \mathbf{1} \otimes \mathbf{1}}_{s \text{ elements}}
\end{aligned}
$$

where $Z = R - \mathrm{E}_{\text{cols}}[R]$, $\mathrm{E}_{\text{cols}}[R]$ is the expected return of each column of returns matrix R, $\mathbf{1}$ is a matrix of ones of size $1 \times n$ and $\odot$ is the Hadamard product operator.

Additionally, we can calculate matrixes Σ_{2k} using the following matricial formulas:

$$
\begin{aligned}
\Sigma_2 &= \frac{1}{T}\left[Z'Z\right] \\
\Sigma_4 &= \frac{1}{T}\left[\left((Z \otimes \mathbf{1}) \odot (\mathbf{1} \otimes Z)\right)'\left((Z \otimes \mathbf{1}) \odot (\mathbf{1} \otimes Z)\right)\right] \\
&\;\;\vdots \\
\Sigma_{2k} &= \frac{1}{T}\left[\left(\bigodot_{i=1}^{k} W_i^k\right)'\left(\bigodot_{i=1}^{k} W_i^k\right)\right]
\end{aligned}
$$

Finally, if we want to calculate lower semi-co-moments matrixes we only have to make $Z = \min(R - \mathrm{E}_{\text{cols}}[R], 0)$ and for upper semi-co-moments matrixes we make $Z = \max(R - \mathrm{E}_{\text{cols}}[R], 0)$.

B Scenario Based Portfolio Optimization Models

B.1 Kurtosis Portfolio Optimization

Here we show how to posed the kurtosis minimization problem based on scenarios:

$$
\begin{aligned}
\min_{x,\, d} \quad & \left(\frac{1}{T} \sum_{t=1}^{T} d_t^4 \right)^{1/4} \\
\text{s.t.} \quad & r_t y - M_1 y \leq d_t \\
& r_t y - M_1 y \geq -d_t \\
& \sum_{i=1}^{n} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall \; i \in N \\
& d_t \geq 0 \; ; \; \forall \; t \in T
\end{aligned}
\tag{18}
$$

where d_t is an auxiliary variable that represents the mean absolute difference in period t and r_t is a row vector that represents assets returns in period t.

In the case of max return risk ratio based on kurtosis, the problem can be posed as follows:

$$
\begin{aligned}
\min_{y,\, d,\, k} \quad & \left(\frac{1}{T} \sum_{t=1}^{T} d_t^4 \right)^{1/4} \\
\text{s.t.} \quad & r_t y - M_1 y \leq d_t \\
& r_t y - M_1 y \geq -d_t \\
& M_1 y - r_f k = 1 \\
& \sum_{i=1}^{n} y_i = k \\
& y_i \geq 0 \; ; \; \forall \; i \in N \\
& d_t \geq 0 \; ; \; \forall \; t \in T
\end{aligned}
\tag{19}
$$

where k is an auxiliary variable and the optimal solution can be obtained making the transformation $x = y / \sum_{i=1}^{n} y_i$.

Finally, the case of risk parity based on kurtosis can be posed as follows:

$$
\begin{aligned}
\min_{y,\, d,\, k} \quad & \left(\frac{1}{T}\sum_{t=1}^{T} d_t^4\right)^{1/4} \\
\text{s.t.} \quad & r_t y - M_1 y \leq d_t \\
& r_t y - M_1 y \geq -d_t \\
& \sum_{i=1}^{n} b \ln(y_i) \geq 1 \\
& \sum_{i=1}^{n} y_i = k \\
& y_i \geq 0 \; ; \; \forall \, i \in N \\
& d_t \geq 0 \; ; \; \forall \, t \in T
\end{aligned} \tag{20}
$$

where k is an auxiliary variable, b is the risk contribution constraint vector and the optimal solution can be obtained making the transformation $x = y / \sum_{i=1}^{n} y_i$.

B.2 Semi-Kurtosis Portfolio Optimization

In the case of semi-kurtosis, we can posed the minimization problem as follows:

$$
\begin{aligned}
\min_{x,\, d} \quad & \left(\frac{1}{T}\sum_{t=1}^{T} d_t^4\right)^{1/4} \\
\text{s.t.} \quad & r_t y - M_1 y \geq -d_t \\
& \sum_{i=1}^{n} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall \, i \in N \\
& d_t \geq 0 \; ; \; \forall \, t \in T
\end{aligned} \tag{21}
$$

where d_t is an auxiliary variable that represents the returns below the mean in period t.

In the case of max return risk ratio based on semi-kurtosis, the problem can be posed as follows:

$$
\begin{aligned}
\min_{y,\,d,\,k} \quad & \left(\frac{1}{T}\sum_{t=1}^{T} d_t^4\right)^{1/4} \\
\text{s.t.} \quad & r_t y - M_1 y \geq -d_t \\
& M_1 y - r_f k = 1 \\
& \sum_{i=1}^{n} y_i = k \\
& y_i \geq 0 \;;\; \forall\, i \in N \\
& d_t \geq 0 \;;\; \forall\, t \in T
\end{aligned}
\tag{22}
$$

where k is an auxiliary variable and the optimal solution can be obtained making the transformation $x = y/\sum_{i=1}^{n} y_i$.

Finally, the case of risk parity based on semi-kurtosis can be posed as follows:

$$
\begin{aligned}
\min_{y,\,d,\,k} \quad & \left(\frac{1}{T}\sum_{t=1}^{T} d_t^4\right)^{1/4} \\
\text{s.t.} \quad & r_t y - M_1 y \geq -d_t \\
& \sum_{i=1}^{n} b \ln(y_i) \geq 1 \\
& \sum_{i=1}^{n} y_i = k \\
& y_i \geq 0 \;;\; \forall\, i \in N \\
& d_t \geq 0 \;;\; \forall\, t \in T
\end{aligned}
\tag{23}
$$

where k is an auxiliary variable, b is the risk contribution constraint vector and the optimal solution can be obtained making the transformation $x = y/\sum_{i=1}^{n} y_i$.

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